

Seat No.	
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Set **P**

**B.Sc. (E.C.S.) (Semester - I) (New) (NEP CBCS) Examination:
March/April - 2025
Numerical Methods (G06-0104)**

Day & Date: Thursday, 08-May-2025
Time: 12:00 PM To 01:30 PM

Max. Marks: 30

Instructions: 1) All questions are compulsory.
2) Figures to right indicate full marks.

Q.1 Choose correct alternative. (MCQ)**06**

- 1) While doing the multiplication of two numbers' in normalized floating point the exponents should be _____.
 - a) added
 - b) subtracted
 - c) divided
 - d) multiplied
- 2) If the data is equally spaced and interpolation is near the end of the data then _____ interpolation formula is used.
 - a) Newton's Forward Difference
 - b) Newton's Backward Difference
 - c) Newton's Divided Difference
 - d) Lagrange's
- 3) The equation $x + y = 4$ is _____ equation.
 - a) linear
 - b) tangent
 - c) transcendental
 - d) none of these
- 4) A matrix having only one column is called as _____.
 - a) Row matrix
 - b) Column matrix
 - c) Scalar matrix
 - d) None of these
- 5) $E^n f(x) =$ _____.
 - a) $f(x)$
 - b) $f(x + nh)$
 - c) $f(x - nh)$
 - d) $f(x + 2nh)$
- 6) $\Delta K =$ _____ K is constant
 - a) K
 - b) 0
 - c) 1
 - d) -1

Q.2 Answer the following. (Any Three)**06**

- a) Prepare the forward difference table for following data:

X	2	4	6	8	10
F(x)	12	17	31	52	69

- b) Find the interval in which root of the equation $x^5 + 5x + 1 = 0$ lies.
 c) Define scalar matrix with example.
 d) Define polynomial equation of degree n.
 e) Evaluate: $0.4399 E_2 \times 0.5789 E_{-12}$

Q.3 Answer the following questions. (Any Two)**06**

- a) Find the root of equation $x^3 - 4x - 9 = 0$ by using bisection method, performing only three interactions.
 b) Evaluate a) $1.4586E_{15} \times 7.3254E_6$ b) $1.215E_4 \div 4.5263E_4$
 c) Evaluate Δx^3 by taking $h = 1$

Q.4 Answer the following questions. (Any Two)**06**

- a) Define absolute error, relative error & percentage error
 b) If $f(1) = 5$, $f(12) = 11$, $f(5) = 29$, Then find $f(4)$ by Lagrange's interpolation formula.
 c) Find the root of equation $x^3 + x - 1 = 0$ by using Regula Falsi method, performing only two interactions.

Q.5 Answer the following questions. (Any One)**06**

- a) Estimate
- $f(12)$
- by Newton's forward difference interpolation formula using following data

x:	10	15	20	25
f(x):	14	18	28	42

- b) Solve following system by Jacobi's Iterative method perform only two interactions.

$$10x + y + z = 12$$

$$2x + 10y + z = 13$$

$$x + y + 5z = 7$$